

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · PURE MATHEMATICS
MATHEMATICS 6042/1
PAPER 1 June 2023
3 hours

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: A calculator may be used unless a question forbids it.

INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. Find the first four terms in the expansion of $(1 + 2x)^7$ in ascending powers of x . **[8]**
2. The polynomial $p(x) = x^3 + ax^2 - 3x + 4$ has remainder 4 when divided by $(x - 1)$. Show this and find $p(-1)$. **[8]**
3. Differentiate $y = 7x^3 - 2x^2 + 3x - 1$. Hence find the gradient at $x = 1$. **[8]**
4. Find $\int (7x^2 - 2x + 3) dx$ and the definite integral from 0 to 1. **[8]**
5. Prove that $1 + \tan^2\theta = \sec^2\theta$. Hence solve $1 + \tan^2\theta = 4$ for $0^\circ \leq \theta \leq 180^\circ$. **[8]**
6. A is (1, 2), B is (4, 7).
 - (a) Midpoint [2]
 - (b) Distance AB [3]
 - (c) Equation of AB [3] **[8]**
7. An AP has first term 2 and common difference 2. Find (a) the 12th term [3] (b) S_{12} [5]. **[8]**
8. Solve $3^{(2x)} = 27$. Also simplify $\log_3 81$. **[8]**
9. $f(x) = 2x - 2$, $g(x) = x^2$.
 - (a) $fg(x)$ [2]
 - (b) $gf(x)$ [3]
 - (c) $f^{-1}(x)$ [3] **[8]**
10. Express $(5x+1)/((x+1)(x-2))$ in partial fractions. (Numbers may be checked by covering.) **[8]**
11. $A = (2, 3)$, $B = (5, 7)$. Find $|AB|$ and a unit vector in the direction of AB. **[8]**
12. A circle has radius 6 cm. A sector has angle 1 radians.
 - (a) Arc length [3]
 - (b) Sector area [5] **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).